

Section 1

1.1 Project introduction & problem statement

Floods are one of the most dangerous natural disasters, causing significant loss of life, property, and environmental resources. After a flood, rescue teams must locate people who may be stranded in isolated or dangerous areas. However, post-flood environments can be difficult to access due to damaged roads, strong currents, debris, unstable structures, and limited visibility. These conditions increase the risk to rescue personnel and can make conventional ground-based search operations slow and difficult.

Recent studies have demonstrated the potential of Unmanned Aerial Vehicles (UAVs) to support search and rescue operations by providing aerial observation over large and difficult-to-access areas. UAVs can collect information from above without requiring rescue personnel to immediately enter hazardous areas. Research on UAV-based disaster monitoring and cooperative navigation has also demonstrated that UAVs can be used to improve area coverage and provide information about disaster-affected environments.

Human detection is an important part of search and rescue operations. However, detecting people manually from drone footage can be difficult and time-consuming, especially when monitoring large post-flood areas. In addition, factors such as low-light conditions, weather, complex backgrounds, and visual obstructions may affect the ability of rescue personnel to identify humans. Therefore, the camera mounted on the UAV can be used to capture images or video from the affected area, while image processing and AI-based human detection techniques can support the automatic identification of people from the captured visual data. This can reduce the need for continuous manual observation and help rescue personnel identify possible human locations more efficiently.

However, detecting a person alone is not sufficient for rescue operations. Rescue personnel also need to know where the detected person is located so that the information can be used for further rescue planning. Therefore, the detected human position can be estimated using the UAV's sensor and positioning information and then transmitted to a ground station or rescue personnel in real time.

Therefore, our team aims to develop a UAV-based system that combines AI-based human detection, thermal/visual sensing, localization, and wireless communication to assist rescue teams in post-flood environments. The primary objective of the system is to detect humans, determine and mark their locations, and transmit the detected information back to rescue personnel. By providing this information from a safe distance, the system can reduce the need for rescuers to immediately enter hazardous areas and support them in making safer and more effective decisions during flood response operations. At this stage, the system focuses on human detection and location reporting, rather than autonomous path planning or approaching the detected person.

1.2 Objectives

1. To develop an AI-based system capable of detecting humans from UAV camera data.
2. To estimate and report the location of detected humans using UAV sensor information.
3. To develop a system capable of processing detection information efficiently.
4. To provide a monitoring interface for displaying UAV and human detection information.
5. To evaluate the performance of the proposed human detection and localization system.

1.3 Scope of the Project

The scope of this project covers the development of a UAV-based human detection and localization system. The system consists of UAV hardware, camera and sensor components, an AI-based detection module, and a monitoring system.

The system focuses on detecting possible humans within the UAV camera field of view and estimating their locations based on available sensor and positioning data.

1.4 Expected Results

The expected result of this project is a prototype system that can detect possible humans from UAV camera data and provide their estimated locations to an operator.

The system is expected to support search and rescue operations by reducing the time required to identify possible human locations and providing useful information for further investigation.